

Presentation on Combinatorial group Theory

I. Introduction

Book: *The History of Combinatorial Group Theory*, by Bruce Chandler and Wilhelm Magnus (NY, USA).

- Combinatorial group theory is characterized as the theory of groups which are given by generators and relations, which are called presentations today.
- But this is NOT the complete definition of the field, and let's NOT try to give one here.
- The presentation is going to be split into 2 parts, very similar to the book.

Part 1: We will talk about the period from 1882–1918, the end of the First World War.

The reason we pick this timeline is because mathematical production was reduced in extent and quantity for the countries involved in the war. It also marked a change in the nature of research in this field.

Part 2: Here, new authors and new problems appeared after the war. We are going to talk about this in part 2.

Part 1

1. Walther von Dyck (German)

The paper by Walther von Dyck in **1882** is the first paper in which generators and defining relations are introduced as new concepts, but later were also used for mathematical research.

The book describes that ever since the report on von Dyck's first paper, it unfolded new concepts, problems, results, and its relation with other fields, specifically topology.

Proposition 1. There exists a group G on m generators $A_1, A_2, \dots, A_m$ such that for all $g \in G$ we can write

$$g = A_{i_1}^{e_1} A_{i_2}^{e_2} \cdots A_{i_\ell}^{e_\ell} \quad (\text{or } 1),$$

where $i_k \in \{1, 2, \dots, m\}$, $i_k \neq i_{k+1}$ for $k = 1, \dots, \ell - 1$, and e_k is a nonzero integer.

Remark. The conditions $i_k \neq i_{k+1}$ (no immediate repetition) and $e_k \neq 0$ prevent writing $A_i^0 = 1$. Such a word is called a **reduced word**.

What group is this describing? It is the free group on m generators, F_m . "Free" means there are no relations among the generators other than the group axioms: closure, associativity, identity, and inverse.

Group action: Concatenate two reduced words, cancel every occurrence of a letter followed by its inverse, and merge exponents. For example,

$$(aab^{-1})(bab^{-1}) = aab^{-1}bab^{-1} = aaab^{-1} = a^3b^{-1}.$$

Uniqueness: "Two distinct reduced words can never be reduced to the same word."

Example of such a group: Take generators $A_1 = a$ and $A_2 = b$. The group is the free group $F_2 = \langle a, b \rangle$. For example, an element is

$$w = a^3b^{-2}a^{-1}a^2bb^{-3}a \longrightarrow a^3b^{-2}ab^{-2}a.$$

Proposition 2. Let $F_m = \langle A_1, A_2, \dots, A_m \rangle$ be the free group on m generators. “Every group $\overline{G}$ that can be generated by m generators,

$$\overline{G} = \langle \overline{A}_1, \dots, \overline{A}_m \rangle,$$

is a homomorphic image (quotient) of F_m .”

In other words, pick any set of reduced words $F_1, F_2, \dots, F_r \in F_m$, and declare them trivial after replacing A_i with $\overline{A}_i$:

$$F_j(\overline{A}_1, \dots, \overline{A}_m) = 1 \quad (j = 1, \dots, r).$$

Then the group presented by the generators $\overline{A}_i$ is the quotient

$$\overline{G} \cong F_m / \langle\langle F_1, \dots, F_r \rangle\rangle,$$

where $\langle\langle \cdot \rangle\rangle$ denotes normal closure.

Normal closure:

$$\langle\langle S \rangle\rangle = \langle Ts^{\pm 1}T^{-1} \mid s \in S, T \in G \rangle.$$

Example: When $m = 2$, let

$$F_1 = A_1^2, \quad F_2 = A_2^2, \quad F_3 = A_2 A_1 A_2^{-1} A_1^{-1}.$$

Then the handwritten example states

$$F_2 / \langle\langle A_1^2, A_2^2, A_2 A_1 A_2^{-1} A_1^{-1} \rangle\rangle \cong \langle a, b \mid a^2 = 1, b^3 = 1, bab^{-1} = a^{-1} \rangle.$$

Transition from von Dyck to Arthur Cayley: “Having shown with Dyck that every group generated by symbols $A_1, \dots, A_m$ is a quotient of the F_m obtained by adding relations. But, we now face a practical question: How can we see these relations at work and keep track of the resulting quotient?” The answer was given by Arthur Cayley after a few years.

2. Arthur Cayley

He proposed drawing directed, coloured graphs, which are called *Cayley graphs* now. Their vertices are elements of the group; their edges record right multiplication by the generators.

Example 1: A cyclic group of order 4 has the presentation

$$C_4 = \langle a \mid a^4 = 1 \rangle, \quad C_4 = \{1, a, a^2, a^3\}.$$

The generator a is blue (solid arrow). Its inverse $a^{-1} = a^3$ is red (dashed arrow).

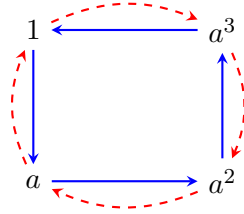
Add directed, coloured edges: for every $g \in C_4$ and each generator $x \in \{a, a^{-1}\}$, draw an edge

$$g \xrightarrow{x} gx.$$

The edge table in the notes is:

From	$x = a$ (blue)	$x = a^{-1}$ (red)
1	$1 \rightarrow a$	$1 \rightarrow a^{-1} \rightarrow a^3$
a	$a \rightarrow a^2$	$a \rightarrow 1$
a^2	$a^2 \rightarrow a^3$	$a^2 \rightarrow a$
a^3	$a^3 \rightarrow a^4 \rightarrow 1$	$a^3 \rightarrow a^2$

Visual: The Cayley graph is a square. Its blue solid edges travel $1 \rightarrow a \rightarrow a^2 \rightarrow a^3 \rightarrow 1$, and the red dashed inverse edges travel in the other direction:



So the closed blue loop of length 4 gives us

$$aaaa = a^4 = 1.$$

This is exactly the defining relator!

Example 2: Start with a free group $F_1 = \langle a \rangle$. Its Cayley graph is an infinite line,

$$\dots \xrightarrow{a} \bullet \xrightarrow{a} \bullet \xrightarrow{a} \bullet \xrightarrow{a} \dots$$

Hence, adding a single relation $a^4 = 1$ quotients that line onto a 4-cycle. This illustrates von Dyck's principle: "A relation corresponds to a closed path to a point."

So, the Cayley graph lets us picture a presentation; what's next?

(3) Henri Poincaré (1895)

Poincaré looks at a space instead of a group and asks:

What algebraic object captures all loops on this surface?

The answer: fundamental group π_1 .

A cell decomposition of the surface immediately turns π_1 into a presentation:

$$\text{cell decomposition of a surface} \longrightarrow \text{presentation}.$$

Hence, topology suddenly needed combinatorial presentations, which is exactly the language Cayley & von invented.

(4) Heinrich Tietze (1908)

Tietze asks a question:

"If two presentations describe the same group, how can I move from one to the other?"

He then discovers reversible Tietze transformations:

- Add or delete a generator.
- Add or delete a redundant relator.

(5) Max Dehn (1910–1914)

Algorithm arrive.

Dehn changes the focus from structure to decision:

- Word problem: is a given word the identity? (Dehn algorithm)
- Conjugacy problem: are two words conjugate?

Interesting fact: Dehn’s connection with Cayley graphs.

Part II

Transition to part II:

“Say” in class: With Poincaré we dive into topology, with Tietze we learn to rewrite, and with Dehn we start computing, although each step is still visible in the Cayley graph. Hence, the language of group theory is now fully pronounceable.

1914–1918 was the period of WWI.

So, as we discussed earlier, mathematical production was grim for the countries involved.

Hence, a new wave of this field has started.

They are no longer satisfied with importing questions from topology, so they ask:

“What theorems are true inside this language itself?”

(1) Jakob Nielsen (1918)

Nielsen transformations:

“Any basis of the free group F_n can be transformed into any other by 3 moves:”

- (i) Invert a generator: $x_i \mapsto x_i^{-1}$.
- (ii) Swap two generators: $x_i \longleftrightarrow x_j$.
- (iii) “Scale”: $x_i \mapsto x_i x_j$ or $x_j x_i$.

He proves these generate $\text{Aut}(F_n)$, the automorphisms (preserving multiplication and having an inverse).

Example in $F_2 = \langle a, b \rangle$:

$$\varphi : \begin{cases} a \mapsto ab, \\ b \mapsto a^{-1}. \end{cases}$$

Let’s do an abelianisation test:

$$\varphi : F_2 \longrightarrow \mathbb{Z}_2, \quad a \mapsto (1, 0), \quad b \mapsto (1, 0).$$

Then $ab \mapsto (1, 1)$ and $a^{-1} \mapsto (-1, 0)$.

In a 2×2 matrix:

$$M = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad \det M = 1.$$

So, in $\mathbb{Z}^2$ a pair of vectors is a basis if and only if its determinant is ± 1 .

Thus, both images form a free basis. Hence, $\varphi \in \text{Aut}(F_2)$.

(2) Reidmeister (1926)–Schreir (1927)

Reidmeister: He formulates a procedure to obtain a presentation of a normal subgroup $H \trianglelefteq G$ of infinite index d given a presentation of G . (Skip)

Otto Schreir: Generalises to arbitrary subgroups and introduces the Schreir transversal lemma.

Interesting/sad fact: Schreir’s parents were killed in the concentration camp during Holocaust, where European Jews were killed.

Schreir’s lemma

Suppose H is a subgroup of G , $H \leq G$, which is finitely generated by S , i.e. $G = \langle S \rangle$.

Let R be the right transversal for the cosets of H in G . (A transversal is a set containing exactly one element from each member of the collection.)

Define, for all $g \in G$, $\bar{g} \in R$ to be the unique element of R with $g\bar{g}^{-1} \in H$.

Then H is generated by the set

$$T = \left\{ rs(\bar{r}\bar{s})^{-1} \mid r \in R, s \in S \right\}.$$

Example: $G = S_3$ and $H = \mathbb{Z}_3$, a subgroup of G .

$$\mathbb{Z}_3 = \mathbb{Z}/3\mathbb{Z}.$$

Thus $\mathbb{Z}_3$ is a subgroup of S_3 :

$$\mathbb{Z}_3 = \{e, (123), (132)\}, \quad S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

Now choose a finite generating set for S_3 :

$$S = \{s_1 = (12), s_2 = (123)\}$$

(because any transposition and any 3-cycle generate S_3).

So, list the right cosets of H in G . By Lagrange’s theorem, $|S_3| = 6$ and $|\mathbb{Z}_3| = 3$. Hence,

$$[S_3 : \mathbb{Z}_3] = \frac{|S_3|}{|\mathbb{Z}_3|} = \frac{6}{3} = 2.$$

Thus there are exactly two right cosets:

- $g = e$ gives $He = H$;
- $g \notin H$, so consider $H(12)$.

$h \in H$	$h(12)$
e	(12)
(123)	$(123)(12) = (13)$
(132)	$(123)(132) = (23)$

Hence, the cosets are H and

$$H(12) = \{(12), (13), (23)\}.$$

Now select the right transversal:

$$R = \{t_1 = e, t_2 = (12)\}.$$

(It contains one representative from each coset.)

Compute rs and choose $\overline{rs} \in R$:

$r \in R$	$s \in S$	rs	Which coset?	$\overline{rs}$
$t_1 = e$	$s_1 = (12)$	(12)	$H(12)$	$t_2 = (12)$
$t_1 = e$	$s_2 = (123)$	(123)	H	$t_1 = e$
$t_2 = (12)$	$s_1 = (12)$	e	H	$t_1 = e$
$t_2 = (12)$	$s_2 = (123)$	(23)	$H(12)$	$t_2 = (12)$

Form the Schreier generators $rs(\overline{rs})^{-1}$:

$$t_1 s_1 (\overline{t_1 s_1})^{-1} = (12)(12)^{-1} = e,$$

$$t_1 s_2 (\overline{t_1 s_2})^{-1} = (123)e^{-1} = (123),$$

$$t_2 s_1 (\overline{t_2 s_1})^{-1} = ee^{-1} = e,$$

$$t_2 s_2 (\overline{t_2 s_2})^{-1} = (23)(12)^{-1} = (132).$$

Thus, Schreier's lemma gives the following generators for $\mathbb{Z}_3$:

$$\{e, (123), (132)\}.$$

Since $(123)^{-1} = (132)$, we can simplify this to

$$\{(123)\}.$$

Therefore, we have seen that Schreier's lemma shows that the subgroup $\mathbb{Z}_3$ is generated by (123) .

Schreier's lemma

Suppose H is a subgroup of G , $H \leq G$, which is finitely generated by S , i.e., $G = \langle S \rangle$. Let R be the right transversal for the cosets of H in G .

For every $g \in G$, denote by $\bar{g} \in R$ the unique element from R with $g\bar{g}^{-1} \in H$. Then H is generated by the set

$$T = \{\gamma s (\overline{\gamma s})^{-1} \mid \gamma \in R, s \in S\}.$$

Proof of Schreier's lemma. (1) We show $T \subseteq H$. Let $\gamma s (\overline{\gamma s})^{-1} \in T$, where $\gamma \in R$ and $s \in S$.

By definition of $\overline{\gamma s}$, we have the identity

$$\gamma s (\overline{\gamma s})^{-1} \in H.$$

Thus, every element of T is explicitly constructed to be an element of H . Therefore $T \subseteq H$; consequently $T^{-1} \subseteq H$.

(2) Let $h \in H$. Since $G = \langle S \rangle$, let

$$h = s_1 s_2 \cdots s_n, \quad s_i \in S \cup S^{-1}, \quad n \geq 0.$$

If $n = 0$, the product is the empty word. Now construct sequences

$$\gamma_1, \gamma_2, \dots, \gamma_{n+1} \in R, \quad u_1, u_2, \dots, u_n \in H.$$

For n letters, we get $n + 1$ representatives. **Base step:** Set $\gamma_1 = 1 \in R$, since the empty prefix lies in H , whose unique representative in R is 1.

Inductive step: Assume $\gamma_j \in R$ has been chosen for some $1 \leq j \leq n$. Define

$$\gamma_{j+1} = \overline{\gamma_j s_j} \in R,$$

which is the unique representative of the right coset $H\gamma_j s_j$. Set

$$u_j = \gamma_j s_j \gamma_{j+1}^{-1}.$$

We show $u_j \in T \cup T^{-1}$.

Case 1: If $s_j \in S$, then

$$u_j = \gamma_j s_j (\overline{\gamma_j s_j})^{-1}.$$

This is exactly the definition of an element of T .

Case 2: If $s_j \in S^{-1}$, write $s_j = t^{-1}$ with $t \in S$. Observe that

$$u_j = \gamma_j t^{-1} \gamma_{j+1}^{-1}, \quad u_j^{-1} = \gamma_{j+1} t \gamma_j^{-1}.$$

We know $\gamma_{j+1} = \overline{\gamma_j s_j} = \overline{\gamma_j t^{-1}}$. Thus γ_{j+1} is the unique element of R in the right coset $H\gamma_j t^{-1}$, i.e.,

$$H\gamma_{j+1} = H\gamma_j t^{-1} \implies H\gamma_{j+1} t = H\gamma_j.$$

This means $\gamma_j = \overline{\gamma_{j+1} t}$, since R contains only one representative of each coset and they must coincide.

Then

$$u_j^{-1} = \gamma_{j+1} t (\overline{\gamma_{j+1} t})^{-1} \in T.$$

Thus $u_j \in T^{-1}$, so $u_j \in T \cup T^{-1} \subseteq H$.

By construction, $\gamma_{j+1} = \overline{\gamma_j s_j}$, and hence

$$\gamma_j s_j = u_j \gamma_{j+1}.$$

As $\gamma_1 = 1$, we obtain

$$\begin{aligned} h &= s_1 s_2 \cdots s_n \\ &= \gamma_1 s_1 s_2 \cdots s_n \\ &= u_1 \gamma_2 s_2 \cdots s_n \\ &= u_1 u_2 \gamma_3 s_3 \cdots s_n \\ &\vdots \\ h &= u_1 u_2 \cdots u_n \gamma_{n+1}. \end{aligned} \tag{\star}$$

Each u_j belongs to $T \cup T^{-1}$. Equation $(\star)$ expresses h as a product of the $u_j \in H$ and $\gamma_{n+1} \in R$. Hence

$$\gamma_{n+1} = (u_1 \cdots u_n)^{-1} h \in H.$$

But $\gamma_{n+1} \in R$ and $H \cap R = \{1\}$, since R intersects each right coset exactly once and $1 \in R$ is the unique element in the coset H . Consequently $\gamma_{n+1} = 1$. Hence

$$h = u_1 u_2 \cdots u_n, \quad u_j \in T \cup T^{-1}.$$

Thus $H \subseteq \langle T \cup T^{-1} \rangle$. By (1), $T \cup T^{-1} \subseteq H$, so $\langle T \cup T^{-1} \rangle \subseteq H$. Therefore

$$H = \langle T \rangle.$$

Conclusion

We have started with Dyck's idea of defining a group by symbols and relations. Postwar, Schreier and Nielsen have discovered the first theorems.